

Laboratory 5

TASKS TO BE CARRIED OUT

Create socket Server with multiple Clients in python (on MS Windows):

1. Create a Server code (host = '127.0.0.1' port = the first 4 digits of your index number). (0.5 points)
2. Create a Client code. Make sure that the Client keeps running as the Server is running. Use **multithreading** programming techniques to handle multiple client connections at the same time. (1 point)
3. Server and Clients should exchange messages: (1 point)
 - The welcome message from the Server = “I am a” + your surname + “Server”.
 - Get input message from Client/user (form keyboard) and send it to the Server.
4. Run the Server and four Clients. If the Server (**or Clients**) sends "closed" in the payload then close the socket. (0.75 points)
5. Transfer text files: 'text1.txt', 'text2.txt' from Client to the Server. Implement buffering techniques to efficiently handle large data transfers. (1 point)
6. Use try-except-finally blocks to catch and handle specific types of errors. Close connections properly to release resources and avoid memory leaks. (0.75 point)

Make **print-screens** of the results and upload them on e-nauczanie. **Upload also the Server and Client code!!!**